

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Develop Dynamic Webpage Using Java Script

2. AIM

- To develop a dynamic, interactive webpage using **HTML, CSS, and JavaScript**.
 - To create a **Dynamic Student Dashboard** that includes a live digital clock, theme toggle functionality, a percentage calculator, a motivational quote generator, and a student registration form with validation.
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3. OBJECTIVE

The objectives of this experiment are:

- To understand the concept of developing dynamic, interactive webpages using JavaScript.
 - To use JavaScript functions and methods to perform real-time operations like clock updates.
 - To dynamically manipulate the DOM to update page content based on user interaction.
 - To implement input validation for the percentage calculator and registration form.
 - To toggle interface themes dynamically using CSS classes.
 - To handle event listeners to trigger dashboard features like quote generation and theme switching.
-

4. THEORY

JavaScript is a programming language used to make webpages dynamic and interactive, allowing content to be created, modified, and removed based on user actions.

In this application, JavaScript is used to develop a **Dynamic Student Dashboard** that helps users manage academic tasks.

- **HTML** provides the structure for the dashboard components (clock, calculator, form, quotes).
- **CSS** provides the styling, including transitions for dark mode.
- **JavaScript functions** manage the logic:
 - **Date()** and **setInterval()** maintain the live clock.
 - **Math.random()** handles the dynamic quote generation.
 - **classList.toggle()** manages the theme switching.
 - **Event listeners** (**addEventListener**) handle form submissions and button interactions.
 - **DOM methods** (**getElementById**, **innerHTML**) access and update elements based on user input.

The basic flow of the application is:

User Interaction → JavaScript Logic (Logic/Validation/Math) → DOM Update (Dynamic UI Changes)

5. ALGORITHM / PROCEDURE

1. Create the HTML structure containing a header, live clock display, theme toggle button, student info section, percentage calculator, motivational quote display, and registration form.
 2. Implement dark mode theme toggle functionality using **classList.toggle()** attached to the theme button event listener.
 3. Create a digital clock function utilizing **Date()** and **setInterval()** to dynamically update and display the current time every second.
 4. Implement a percentage calculator that validates input marks to ensure they fall within the range of 0 to 100, then computes and displays the average percentage.
 5. Create a quote generator function that selects and displays random motivational quotes from a predefined array upon button click.
 6. Implement form validation on submission that checks for empty required fields and validates email format before confirming registration.
 7. Apply CSS rules to layout and style page components, and define properties for dark mode styling.
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6. PROGRAM

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Dynamic Student Dashboard</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1 id="heading">Dynamic Student Dashboard</h1>
    <p id="WelcomeMessage">Welcome to the Dynamic Student
Dashboard! This dashboard provides an interactive
    experience for students to manage their academic
activities and track their progress.</p>

    <section class="clock">
      <h2>Current Time</h2>
      <h3 id="clock">00:00:00</h3>
    </section>

    <section>
      <button id="themeBtn">Toggle Theme</button>
    </section>

    <section class="student">
      <h2>Student Information</h2>

      <p>Name : Bhuvan</p>

      <p>Department : AIML</p>

      <p>Year : 3rd Year</p>

    </section>
```

```
<section class="calculator">

    <h2>Percentage Calculator</h2>

    <input type="number" id="m1" placeholder="Subject 1">

    <input type="number" id="m2" placeholder="Subject 2">

    <input type="number" id="m3" placeholder="Subject 3">

    <button id="calcBtn">

        Calculate

    </button>

    <h3 id="result">

        Percentage :

    </h3>

</section>

<section>

    <h2>Motivation</h2>

    <p id="quote">

        Click below for motivation.

    </p>

    <button id="quoteBtn">

        New Quote

    </button>

</section>
```

```
<section>

    <h2>Registration Form</h2>

    <form id="studentForm">

        <input
            type="text"
            id="name"
            placeholder="Name">

        <input
            type="email"
            id="email"
            placeholder="Email">

        <input
            type="text"
            id="roll"
            placeholder="Roll Number">

        <button type="submit">
            Submit
        </button>

    </form>

    <p id="formMessage"></p>

</section>

<footer>

    <p id="footer">

        ©2026

    </p>

</footer>
</header>
```

```
<script src="script.js"></script>
</body>
</html>
```

Script js :

```
let themeBtn = document.getElementById("themeBtn");

let clock = document.getElementById("clock");

let calcBtn = document.getElementById("calcBtn");

let mark1 = document.getElementById("m1");
let mark2 = document.getElementById("m2");
let mark3 = document.getElementById("m3");

let result = document.getElementById("result");

let quoteBtn = document.getElementById("quoteBtn");

let quote = document.getElementById("quote");

let form = document.getElementById("studentForm");

let nameInput = document.getElementById("name");

let emailInput = document.getElementById("email");

let rollInput = document.getElementById("roll");

let formMessage = document.getElementById("formMessage");
```

```
themeBtn.addEventListener("click", function() {  
  
    document.body.classList.toggle("dark-mode");  
  
});
```

```
function updateClock() {  
  
    let currentTime = new Date();  
  
    let hours = currentTime.getHours();  
  
    let minutes = currentTime.getMinutes();  
  
    let seconds = currentTime.getSeconds();  
  
    hours = String(hours).padStart(2, "0");  
  
    minutes = String(minutes).padStart(2, "0");  
  
    seconds = String(seconds).padStart(2, "0");  
  
    clock.innerHTML =  
        hours + ":" + minutes + ":" + seconds;  
  
}
```

```
updateClock();
```

```
setInterval(updateClock, 1000);
```

```
calcBtn.addEventListener("click", function() {  
  
    let marks1 = parseFloat(mark1.value);  
  
    let marks2 = parseFloat(mark2.value);
```

```
let marks3 = parseFloat(mark3.value);

if (
  isNaN(marks1) ||
  isNaN(marks2) ||
  isNaN(marks3)
) {

  result.innerHTML =
    "Please enter all marks.";

  return;
}

if (
  marks1 < 0 || marks1 > 100 ||
  marks2 < 0 || marks2 > 100 ||
  marks3 < 0 || marks3 > 100
) {

  result.innerHTML =
    "Marks must be between 0 and 100.";

  return;
}

let total =
  marks1 + marks2 + marks3;

let percentage =
  (total / 300) * 100;
```



```
result.innerHTML =  
    "Percentage: " +  
    percentage.toFixed(2) +  
    "%";  
  
});  
  
let quotes = [  
  
    "Success comes from consistent effort.",  
  
    "Believe in yourself and keep moving forward.",  
  
    "Small progress is still progress.",  
  
    "Don't stop when you are tired. Stop when you are done.",  
  
    "Your future depends on what you do today.",  
  
    "Great things take time.",  
  
    "Learn from yesterday and build for tomorrow."  
  
];  
  
function generateQuote() {  
  
    let randomIndex =  
        Math.floor(Math.random() * quotes.length);  
  
    quote.innerHTML =  
        quotes[randomIndex];  
  
}  
  
quoteBtn.addEventListener("click", generateQuote);
```

```
generateQuote();

form.addEventListener("submit", function(event) {

    event.preventDefault();

    let name = nameInput.value.trim();

    let email = emailInput.value.trim();

    let roll = rollInput.value.trim();

    if (name === "") {

        formMessage.innerHTML =
            "Please enter your name.";

        return;
    }

    if (
        !email.includes("@") ||
        !email.includes(".")
    ) {

        formMessage.innerHTML =
            "Please enter a valid email.";

        return;
    }
}
```

```
if (roll === "") {

    formMessage.innerHTML =
        "Please enter your roll number.";

    return;
}

if (isNaN(roll)) {

    formMessage.innerHTML =
        "Roll number must contain numbers only.";

    return;
}

formMessage.innerHTML =
    "Registration successful!";

form.reset();

});
```

Style.css:

```
/* Reset */
```

```
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}

/* Body */

body{

    font-family:Arial, Helvetica, sans-serif;

    background:#f5f5f5;

    color:#333;

    padding:30px;

    transition:0.4s;
}

/* Header */

header{

    text-align:center;

    margin-bottom:30px;
}

header h1{

    color:#0d6efd;

    margin-bottom:10px;
}

header p{

    font-size:20px;
}
```

```
/* Sections */

section{

    background:white;

    padding:20px;

    margin:20px auto;

    border-radius:12px;

    width:80%;

    box-shadow:0 0 10px rgba(0,0,0,0.15);
}
```

```
/* Headings */

section h2{

    margin-bottom:15px;

    color:#0d6efd;
}
```

```
/* Inputs */

input{

    width:100%;

    padding:12px;

    margin:10px 0;

    border-radius:8px;

    border:1px solid gray;

    font-size:16px;
```

```
}

/* Buttons */

button{

    background:#0d6efd;

    color:white;

    border:none;

    padding:12px 20px;

    border-radius:8px;

    cursor:pointer;

    margin-top:10px;

    transition:0.3s;
}

button:hover{

    background:#084298;
}

/* Result */

#result{

    margin-top:20px;

    color:green;
}

/* Quote */

#quote{

    font-style:italic;
```

```
        margin-bottom:15px;
    }

    /* Footer */

    footer{

        text-align:center;

        margin-top:30px;

        font-size:18px;
    }

    /* Clock */

    .clock{

        text-align:center;
    }

    #clock{

        font-size:35px;

        color:red;
    }

    /* Student Details */

    .student p{

        margin:10px 0;

        font-size:18px;
    }

    .dark-mode {

        background: #222;
    }
}
```

```
    color: white;
}

.dark-mode section {

    background: #333;

    color: white;
}

.dark-mode section h2 {

    color: #66aaff;
}

.dark-mode header h1 {

    color: #66aaff;
}

#formMessage {

    margin-top: 15px;

    font-weight: bold;

    text-align: center;
}
```

7. CODE EXPLANATION

Component	Description
themeBtn	Button to toggle between light and dark themes.
clock	Displays the current time, updated every second.
calcBtn	Calculates the percentage based on input marks from three subjects.
quoteBtn	Button to display a random motivational quote.
studentForm	Form to capture student name, email, and roll number.
form.addEventListener("submit")	Handles form submission and validates input data.

8. TEST CASES AND EXECUTION DATA

Test Case	Test Scenario	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Load	Open HTML file	Page loads with elements	Page loaded correctly	Pass
TC-02	Theme Toggle	Click "Toggle Theme" button	Theme switches to dark mode	Theme switched	Pass
TC-03	Calculator	Enter valid marks and click "Calculate"	Percentage is displayed	Percentage calculated	Pass
TC-04	Quote Generator	Click "New Quote" button	A new quote appears	Quote updated	Pass
TC-05	Form Validation	Submit form with empty fields	Error message appears	Error displayed	Pass

9. OUTPUT

Dynamic Student Dashboard

Welcome to the Dynamic Student Dashboard! This dashboard provides an interactive experience for students to manage their academic activities and track their progress.

Current Time

23:54:58

Toggle Theme

Student Information

Name : Bhuvan

Department : AIML

Year : 3rd Year

Percentage Calculator

80

98

Calculate

Percentage: 88.67%

Motivation

Don't stop when you are tired. Stop when you are done.

New Quote

Registration Form

Name

Email

Roll Number

Submit

Registration successful!

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10. RESULT

Thus, a **dynamic webpage** was successfully developed using **HTML, CSS, and JavaScript**. The application allows users to dynamically add tasks, mark tasks as completed, and delete tasks. JavaScript DOM manipulation and functions were successfully implemented, and the application was verified using multiple test cases.